Big science and peer review, a game theoretical model

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1 Introduction

2 Model

2.1 Setting

We consider a publication game involving a finite population of N authors and a single academic journal. Our framework aligns with established game-theoretic models of scientific publishing Thurner and Hanel (2011), while specifically building on the incentive structures analyzed by Zollman et al. Zollman et al. (2024). Unlike previous static models, we extend the analysis to account for science volume. The fundamental unit of the game is a research paper, which possesses a binary intrinsic quality $q \in \{0, 1\}$. A paper is defined as "good" ($q = 1$) or "bad" ($q = 0$). In each period, authors make submission decisions, and the journal enforces quality control standards subject to a physical capacity constraint.

The publication process unfolds in a strictly sequential timeline, as illustrated in Figure 1:

1. **Submission (Generation of V):** Based on their strategies, authors decide whether to submit their work. The aggregate collection of all submitted manuscripts forms the *Submission Pool*, denoted by $V \leq N$.

2. **Peer Review (Generation of M):** The journal applies a peer review process to filter the submission pool. Acknowledging the inherent unreliability of assessment documented in empirical studies Bornmann (2011), we model reviewers as imperfect classifiers. They are subject to Type I errors (False Positives, rate λ) and Type II errors (False Negatives, rate ϵ). Papers that successfully pass this screening form the set of *Peer-accepted papers*, denoted by M. It is important to note that membership in M signifies that a paper is "scientifically acceptable," but not necessarily "published."

3. **Capacity Check and Storage (K):** The journal has a fixed resource limit or *Journal Storage*, denoted by K. This constraint reflects the physical limits on publication slots or the administrative burden of processing manuscripts Kovanis et al. (2016).

4. **Final Acceptance (ρ):** The final publication decision depends on the relationship between the volume of peer-accepted papers (M) and the journal storage (K). We define ρ as the *probability of being accepted conditional on being peer-accepted*. This variable captures the congestion effect:

$$\rho = \begin{cases} 1 & \text{if } M \leq K \quad \text{(No Congestion)} \\ K/M & \text{if } M > K \quad \text{(Rationing triggered)} \end{cases} \tag{1}$$

In this framework, a paper is published if and only if it passes two hurdles: the quality hurdle (peer review) and the congestion hurdle (storage constraint).

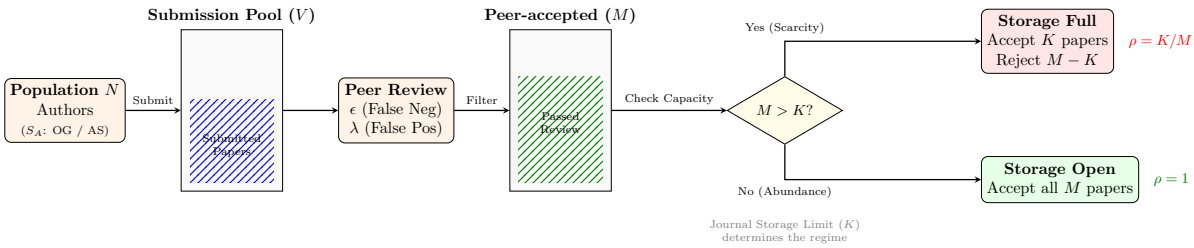


Figure 1: **Schematic representation of the Threshold Competition Model.** The process unfolds in stages: (1) Authors from population N submit papers into pool V; (2) Peer review filters submissions into a set of peer-accepted papers M; (3) The volume M is compared against the journal storage capacity K. If $M > K$, a rationing mechanism is triggered, reducing the conditional acceptance probability ρ below 1.

2.2 Players, strategies, and quality generation

The publication game involves a single journal (J) and a finite population of authors $\mathcal{A} = \{1, 2, \ldots, N\}$. We treat N as a finite exogenous parameter. This finite-population approach allows us to capture the congestion dynamics and "tragedy of the commons" effects inherent in resource-constrained peer review systems, as analyzed in the economic literature on academic publishing (Cotton, 2013; Azar, 2005). Throughout, we use V and M to denote the realised *counts* of submitted and peer-accepted papers, respectively.

Quality generation. In each round, each author generates a manuscript with binary intrinsic quality $q \in \{0, 1\}$, where $q = 1$ denotes a "good" paper and $q = 0$ a "bad" paper. Quality is generated i.i.d. across authors with

$$\Pr(q = 1) = \alpha, \qquad \Pr(q = 0) = 1 - \alpha, \tag{2}$$

where $\alpha \in (0, 1)$ is an exogenous productivity parameter reflecting the baseline scientific capability of the community (Ellison, 2002).

Author strategies (S_A). Each author chooses a submission rule $s_A \in S_A$ that conditions on realised quality. Consistent with signaling models in scientific communication (Tiokhin et al., 2021), we consider four pure strategies:

- **Only Good (OG):** submit iff $q = 1$;
- **Always Submit (AS):** submit regardless of q (strategic spamming);
- **Only Bad (OB):** submit iff $q = 0$ (included for completeness);
- **Never Submit (NS):** never submit.

Equivalently, s_A can be represented by (p_g, p_b), the submission probabilities for good and bad papers:

$$(p_g, p_b) = \begin{cases} (1, 0) & \text{if } s_A = \text{OG} \\ (1, 1) & \text{if } s_A = \text{AS} \\ (0, 1) & \text{if } s_A = \text{OB} \\ (0, 0) & \text{if } s_A = \text{NS}. \end{cases} \tag{3}$$

Journal strategies (S_J). The journal commits to an editorial policy $s_J \in S_J$ that maps the perceived quality signal $\hat{q} \in \{0, 1\}$ into an accept/reject decision:

- **Only Good (OG):** accept iff $\hat{q} = 1$;
- **Always Accept (AA):** accept all submissions (ignores $\hat{q}$);
- **Only Bad (OB):** accept iff $\hat{q} = 0$;
- **Reject All (AR):** reject all submissions (ignores $\hat{q}$).

Exogenous parameters and endogenous variables. To clarify the mechanics of congestion, we distinguish between fixed environmental constraints and emergent outcomes:

- **Exogenous parameters:** Population size (N), productivity (α), journal size (K), review noise parameters (ϵ, λ), and payoff primitives, discussed below: c, cost of submission for authors; r, the reward for an author derived from successful publication; B, the benefit to the journal derived from publishing good papers; and D, the the penalty for journals from publishing bad papers.
- **Endogenous variables:** Submission volume (V), peer-accepted volume (M), the rationing factor ($\rho = \max\left\{\frac{K}{M}, 1\right\}$), and the population state x (strategy shares) governing the equilibrium distribution.

2.3 Review noise and acceptance probabilities

Peer review acts as a stochastic filter between the submission volume V and the peer-accepted volume M. Throughout, *pass* refers to passing the review filter and entering the peer-accepted pool M (not final publication, which is also affected by the journal size.

Noise parameters (ϵ, λ). Consistent with empirical meta-analyses on review reliability (Bornmann et al., 2010; Cicchetti, 1991), we assume reviewers are imperfect classifiers. Let $\hat{q} \in \{0, 1\}$ denote the perceived quality signal for a paper with true quality $q \in \{0, 1\}$. We define:

$$\epsilon = \Pr(\hat{q} = 0 \mid q = 1), \qquad \lambda = \Pr(\hat{q} = 1 \mid q = 0), \tag{4}$$

where ϵ is the false negative rate (acting as a tax on honesty) and λ the false positive rate (acting as an incentive for opportunism), a structure supported by empirical studies on scientific gatekeeping (Siler et al., 2015).

Conditional acceptance probabilities. Given a journal policy s_J, define the probability of acceptance conditional on true quality as:

$$\pi_q(s_J) \equiv \Pr(\text{pass} \mid q, s_J), \qquad q \in \{0, 1\}. \tag{5}$$

Under the journal strategies defined in Section 2.2, and aligned with theoretical models of gatekeeper selection (Muller-Itten, 2022), the acceptance vector (π_1, π_0) is given by:

$$(\pi_1(s_J), \pi_0(s_J)) = \begin{cases} (1-\epsilon,\ \lambda) & \text{if } s_J = \text{OG} \\ (1,\ 1) & \text{if } s_J = \text{AA} \\ (\epsilon,\ 1-\lambda) & \text{if } s_J = \text{OB} \\ (0,\ 0) & \text{if } s_J = \text{AR}. \end{cases} \tag{6}$$

Per-author pass-review probability ($\mathcal{P}$). Let $(p_g(s_A), p_b(s_A))$ denote the submission probabilities conditional on realised quality. We define $\mathcal{P}(s_A, s_J)$ as the *ex-ante* probability that an author playing s_A contributes a paper to the peer-accepted pool M under journal policy s_J:

$$\mathcal{P}(s_A, s_J) = \alpha\, p_g(s_A)\, \pi_1(s_J) + (1-\alpha)\, p_b(s_A)\, \pi_0(s_J). \tag{7}$$

Below we enumerate the effective acceptance probabilities for all author strategies under each journal regime.

Case 1: Standard Review (Journal plays Only Good). This is the baseline scenario where $\pi_1 = 1 - \epsilon$ and $\pi_0 = \lambda$.

$$\mathcal{P}(\text{OG}, \text{OG}) = \alpha(1-\epsilon), \tag{8a}$$
$$\mathcal{P}(\text{AS}, \text{OG}) = \alpha(1-\epsilon) + (1-\alpha)\lambda, \tag{8b}$$
$$\mathcal{P}(\text{OB}, \text{OG}) = (1-\alpha)\lambda, \tag{8c}$$
$$\mathcal{P}(\text{NS}, \text{OG}) = 0. \tag{8d}$$

Case 2: Permissive (Journal plays Always Accept). The screening is ignored ($\pi_1 = 1, \pi_0 = 1$), making acceptance depend solely on submission.

$$\mathcal{P}(\text{OG}, \text{AA}) = \alpha, \tag{9a}$$
$$\mathcal{P}(\text{AS}, \text{AA}) = \alpha + (1 - \alpha) = 1, \tag{9b}$$
$$\mathcal{P}(\text{OB}, \text{AA}) = 1 - \alpha, \tag{9c}$$
$$\mathcal{P}(\text{NS}, \text{AA}) = 0. \tag{9d}$$

Case 3: Inverse Selection (Journal plays Only Bad). The journal accepts papers perceived as bad ($\hat{q} = 0$). Good papers pass only via false negatives ($\pi_1 = \epsilon$), and bad papers pass via true negatives ($\pi_0 = 1 - \lambda$).

$$\mathcal{P}(\text{OG}, \text{OB}) = \alpha\epsilon, \tag{10a}$$
$$\mathcal{P}(\text{AS}, \text{OB}) = \alpha\epsilon + (1 - \alpha)(1 - \lambda), \tag{10b}$$
$$\mathcal{P}(\text{OB}, \text{OB}) = (1 - \alpha)(1 - \lambda), \tag{10c}$$
$$\mathcal{P}(\text{NS}, \text{OB}) = 0. \tag{10d}$$

Case 4: Closed (Journal plays Reject All). No papers enter the peer-accepted pool ($\pi_1 = 0, \pi_0 = 0$).

$$\mathcal{P}(s_A, \text{AR}) = 0 \quad \forall s_A \in S_A. \tag{11}$$

Aggregate peer-accepted volume ($E[M]$). To determine the system-wide congestion, we aggregate these individual passing probabilities across the entire population, a method consistent with burden estimation in peer review (Kovanis et al., 2016). Let $\mathbf{x} = \{x_{s_A} \mid s_A \in S_A\}$ denote the **population state**, where $x_{s_A} \in [0, 1]$ represents the proportion of authors adopting strategy s_A, satisfying $\sum_{s_A \in S_A} x_{s_A} = 1$.

The expected total volume of peer-accepted papers, $E[M]$, is the sum of contributions from all strategy groups:

$$E[M] = \sum_{i=1}^{N} \mathcal{P}(s_{A,i}, s_J) = N \sum_{s_A \in S_A} x_{s_A} \mathcal{P}(s_A, s_J). \tag{12}$$

This aggregate volume $E[M]$ is the critical endogenous variable that interacts with the journal capacity K to determine the rationing probability ρ defined in Eq. (1).

2.4 Capacity constraint and rationing probability (ρ)

A key feature of the model is the separation between *scientific merit* (passing peer review and entering the set of peer-accepted papers) and *publication slots* (capacity-constrained publication). This distinction captures the reality that top-tier journals often reject scientifically sound papers solely due to space limitations (Ellison, 2002; Casadevall and Fang, 2014).

Journal capacity (K). Let $K \in \mathbb{N}^+$ denote the journal's exogenous publication capacity: at most K papers can be published per cycle due to limited pages and editorial resources (Kovanis et al., 2016; Azar, 2005).

Uniform rationing and the rationing probability (ρ). Let M be the realised number of peer-accepted papers in a given round. If $M \leq K$, all peer-accepted papers are published. If $M > K$, the journal applies *uniform rationing*: it selects exactly K papers uniformly at random from the M peer-accepted papers. Hence, the conditional publication probability for a peer-accepted paper is determined by the realised congestion:

$$\rho(M) \equiv \Pr(\text{published} \mid \text{pass}) = \min\left(1, \frac{K}{M}\right) = \begin{cases} 1 & \text{if } M \leq K, \\ \frac{K}{M} & \text{if } M > K. \end{cases} \tag{13}$$

Ex-post vs. ex-ante congestion. Across rounds, the peer-accepted count M is stochastic because it aggregates individual submission and review outcomes. Authors therefore form expectations over congestion given the population state $\mathbf{x}$ and journal policy s_J, implying an expected rationing factor:

$$\mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \frac{K}{M}\right)\right]. \tag{14}$$

Crucially, increases in low-quality or opportunistic submissions raise the expected volume $\mathbb{E}[M]$ and mechanically reduce $\mathbb{E}[\rho]$, generating a negative congestion externality on all authors. This mirrors the inefficiencies found in scientific funding competitions, where excessive participation degrades the system's overall value (Cotton, 2013; Gross and Bergstrom, 2019).

2.5 Payoff analysis

In this subsection, we derive the expected payoffs for both authors and the journal. The introduction of the capacity constraint K fundamentally couples the payoffs: an individual's utility depends not only on their own strategy but also on the aggregate congestion generated by the population. This structure formally aligns with the class of *congestion games*, where individual payoffs are negatively correlated with the total utilization of a shared resource (Rosenthal, 1973).

2.5.1 Notation and general framework

Table 1 summarizes the key parameters and variables. We employ a *mean-field approximation* where players base their decisions on the **expected** state of the system. This approach is standard in evolutionary game theory for modeling interactions in large finite populations, allowing deterministic analysis of stochastic processes (Sandholm, 2010; Weibull, 1995).

Table 1: Summary of key parameters and variables used in the payoff analysis.

Symbol	Description
Exogenous Parameters (Fixed)	
N	Total number of authors
K	Journal capacity (maximum publication slots per cycle)
α	Probability that an author produces a good paper
ϵ	False negative rate (Prob. of rejecting a good paper)
λ	False positive rate (Prob. of accepting a bad paper)
r	Reward for a published paper (Author)
c	Cost of submission (Author)
B	Benefit for publishing a good paper (Journal)
D	Penalty for publishing a bad paper (Journal)
C_J	Unit cost of peer review (per submission)
Endogenous Variables (Dependent on Strategy)	
V	Total number of submissions (Submission Pool)
M	Number of papers passing peer review (Perception Pool)
Y	Number of papers finally published ($Y \leq K$)
$\bar{\rho}$	Expected rationing probability ($\mathbb{E}[\min(1, K/M)]$)
x	Proportion of authors adopting a specific strategy (Population State)

Submission and publication probabilities. For an author playing strategy $s_A \in S_A$, we define the unconditional probability of submitting a paper as $P_{\text{sub}}(s_A)$. Based on the submission vectors (p_g, p_b)

defined in Section 2.2, this is:

$$P_{\text{sub}}(s_A) = \alpha p_g(s_A) + (1-\alpha)p_b(s_A). \tag{15}$$

Recall from Section 2.3 that $\mathcal{P}(s_A, s_J)$ denotes the probability of *entering the peer-accepted pool* M. However, entering M does not guarantee publication. Let $\bar{\rho}$ denote the **expected rationing probability**:

$$\bar{\rho} \equiv \mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \frac{K}{M}\right)\right]. \tag{16}$$

The joint probability of a paper being finally published is the product of passing review and passing the capacity check: $\bar{\rho} \cdot \mathcal{P}(s_A, s_J)$.

Volumes: Submitted (V), Passed (M), and Published (Y). It is crucial to distinguish between papers that pass peer review (M) and those that are finally published (Y). Let V_G and V_B be the expected volumes of good and bad submissions generated by the population. Let M_G and M_B be the expected volumes of papers entering the peer-accepted pool. We define Y_G and Y_B as the expected volumes of **finally published** papers:

$$Y_G = \bar{\rho} M_G, \quad Y_B = \bar{\rho} M_B. \tag{17}$$

2.5.2 General payoff functions

Author's utility (U_A). Let r be the reward for a published paper and c be the cost of submission. The expected payoff for an author playing s_A against a journal s_J is the expected reward from publication minus the reliable cost of submission:

$$U_A(s_A, s_J) = r \cdot \underbrace{\bar{\rho} \cdot \mathcal{P}(s_A, s_J)}_{\text{Prob(Published)}} - c \cdot \underbrace{P_{\text{sub}}(s_A)}_{\text{Prob(Submitted)}} \tag{18}$$

Journal's utility (U_J). Let B be the benefit of publishing a good paper, D be the penalty for a bad paper, and C_J be the unit cost of reviewing a submission. The journal's utility depends on the *published* volumes (Y_G, Y_B) and the *submitted* volume ($V = V_G + V_B$). This formulation captures the trade-off between quality assurance and processing costs (Muller-Itten, 2022):

$$U_J(s_J) = B \cdot Y_G - D \cdot Y_B - C_J \cdot V \tag{19}$$

Substituting $Y = \bar{\rho} M$, we obtain the working form of the journal's payoff:

$$U_J(s_J) = \bar{\rho}\left[B \cdot M_G(s_J) - D \cdot M_B(s_J)\right] - C_J \cdot V \tag{20}$$

Remark: Eq. (20) relies on the approximation $\mathbb{E}[\rho M] \approx \mathbb{E}[\rho]\mathbb{E}[M]$. As N grows large, the relative fluctuations in M diminish, making this mean-field limit exact (Sandholm, 2010).

2.5.3 Derivation of strategy-specific payoffs

We now explicitly list the payoff functions for the 8 primary scenarios (4 Author strategies and 4 Journal strategies).

Author payoffs (under Journal Strategy s_J = OG). In the standard academic context, the journal plays **Only Good (OG)**. The review filter applies $(\pi_1, \pi_0) = (1-\epsilon, \lambda)$. Substituting the specific $\mathcal{P}$ and P_{sub} values into Eq. (18), we derive:

$$U_A(\text{OG}, \text{OG}) = \alpha\left[r\bar{\rho}(1-\epsilon) - c\right] \tag{21}$$

$$U_A(\text{AS}, \text{OG}) = r\bar{\rho}\left[\alpha(1-\epsilon) + (1-\alpha)\lambda\right] - c \tag{22}$$

$$U_A(\text{OB}, \text{OG}) = (1-\alpha)\left[r\bar{\rho}\lambda - c\right] \tag{23}$$

$$U_A(\text{NS}, \text{OG}) = 0 \tag{24}$$

Journal payoffs (under Population State x). The journal chooses a strategy s_J to maximize utility given the incoming flow of submissions. Let $V_G(\mathbf{x})$ and $V_B(\mathbf{x})$ be the volumes of good and bad submissions determined by the population state.

1. Only Good (OG): The journal filters papers. The passed volumes are $M_G = V_G(1-\epsilon)$ and $M_B = V_B\lambda$.

$$U_J(\text{OG}) = \bar{\rho}\,[BV_G(1-\epsilon) - DV_B\lambda] - C_J(V_G + V_B) \tag{25}$$

2. Always Accept (AA): The journal accepts everything ($M_G = V_G, M_B = V_B$). Congestion is maximized here, so $\bar{\rho}$ will be minimal.

$$U_J(\text{AA}) = \bar{\rho}\,[BV_G - DV_B] - C_J(V_G + V_B) \tag{26}$$

3. Only Bad (OB): The journal applies inverse selection ($M_G = V_G\epsilon, M_B = V_B(1-\lambda)$).

$$U_J(\text{OB}) = \bar{\rho}\,[BV_G\epsilon - DV_B(1-\lambda)] - C_J(V_G + V_B) \tag{27}$$

4. Reject All (AR): The journal publishes nothing ($Y = 0$). Review costs are still incurred for processed submissions.

$$U_J(\text{AR}) = -C_J(V_G + V_B) \quad \text{(normalized to 0 in dominance analysis)} \tag{28}$$

2.5.4 Summary of transition probabilities

For completeness, Table 2 summarizes the effective probability $\mathcal{P}(s_A, s_J)$ of entering the peer-accepted pool for any strategy pair. These values are the key inputs for the payoff equations.

Table 2: Matrix of Effective Passing Probabilities $\mathcal{P}(s_A, s_J)$. This represents the probability that an author's paper enters the peer-accepted pool M (before rationing).

Author \ Journal	**Only Good (OG)**	**Always Accept (AA)**	**Only Bad (OB)**	**Reject All (AR)**
OG (α good)	$\alpha(1-\epsilon)$	α	$\alpha\epsilon$	0
AS (All)	$\alpha(1-\epsilon) + (1-\alpha)\lambda$	1	$\alpha\epsilon + (1-\alpha)(1-\lambda)$	0
OB ($1-\alpha$ bad)	$(1-\alpha)\lambda$	$1-\alpha$	$(1-\alpha)(1-\lambda)$	0
NS (None)	0	0	0	0

2.6 Dominance analysis and reduced game

Before analyzing the evolutionary dynamics, we simplify the strategy space by iteratively eliminating strictly dominated strategies. This procedure, known as *iterated elimination of strictly dominated strategies* (IESDS), allows us to focus on the rationalizable core of the game (Fudenberg and Tirole, 1991). We conduct this analysis under the standard **mean-field approximation**, holding the aggregate congestion factor $\bar{\rho}$ fixed. The elimination proceeds in two stages: first, we identify strategies that are dominated purely by the structural logic of the game (independent of specific cost-benefit inequalities); second, we introduce parameter constraints to eliminate strategies based on economic trade-offs.

2.6.1 Stage 1: Structural elimination (Unconditional dominance)

We first establish dominance relations that hold under weak axiomatic assumptions regarding the nature of peer review and scientific value, consistent with signaling models in scientific communication (Tiokhin et al., 2021).

Elimination of Journal "Only Bad" (OB). The strategy **Only Bad (OB)** represents an inverse selection policy. We compare it to **Only Good (OG)**. Using the payoff functions derived in Section 2.5, the difference in expected utility is:

$$\begin{aligned}\Delta U_J &= U_J(\text{OG}) - U_J(\text{OB}) \\ &= \bar{\rho}\,[BV_G(1-\epsilon) - DV_B\lambda] - \bar{\rho}\,[BV_G\epsilon - DV_B(1-\lambda)] \\ &= \bar{\rho}\,[BV_G(1-2\epsilon) + DV_B(1-2\lambda)]\end{aligned} \tag{29}$$

Condition: We assume peer review is **informative**, defined as performing better than random guessing:

$$\epsilon < 0.5 \quad \text{and} \quad \lambda < 0.5. \tag{30}$$

Under this definition, both $(1 - 2\epsilon)$ and $(1 - 2\lambda)$ are strictly positive. Since $B, D, V_G, \bar{\rho} > 0$, it follows that $\Delta U_J > 0$. Thus, **Journal OB is strictly dominated by OG** solely by virtue of the definition of peer review quality, regardless of the magnitude of rewards or penalties.

Elimination of Author "Only Bad" (OB). Consider the choice between **Always Submit (AS)** and **Only Bad (OB)**. AS submits both paper types ($q \in \{0, 1\}$), while OB submits only bad papers ($q = 0$). The payoff difference is the marginal utility of submitting a good paper:

$$\begin{aligned} \Delta U_A &= U_A(\text{AS}, \text{OG}) - U_A(\text{OB}, \text{OG}) \\ &= [U_A(q = 1) + U_A(q = 0)] - U_A(q = 0) \\ &= U_A(q = 1) = \alpha\,[r\bar{\rho}(1 - \epsilon) - c] \end{aligned} \tag{31}$$

Condition: We assume the **Axiom of Value**: in a functional scientific system, a correctly identified high-quality paper yields a positive net return (i.e., $r(1 - \epsilon) > c$). If this axiom holds, then $\Delta U_A > 0$ for any non-zero congestion $\bar{\rho}$. Thus, **Author OB is strictly dominated by AS** due to monotonicity: a rational author will never withhold a high-quality paper while simultaneously submitting a low-quality one.

2.6.2 Parameter constraints for a non-trivial game

Having eliminated the structurally irrational strategies, we now focus on the remaining set. To ensure the game yields interesting evolutionary dynamics (avoiding trivial collapse or utopian perfection), we define the "viable region" of the parameter space using the **abundance benchmark** ($\bar{\rho} = 1$).

1. **Viability Constraint (Honesty is profitable):**

$$r(1 - \epsilon) > c \tag{32}$$

   This formally restates the Axiom of Value. It ensures that without congestion, the honest strategy (OG) dominates silence (NS), a necessary condition for any scientific activity (Gross and Bergstrom, 2019).

2. **Profitability Constraint (Spamming is profitable):**

$$\lambda r > c \tag{33}$$

   This constraint drives the congestion dilemma. It implies that in a frictionless system ($\bar{\rho} = 1$), the expected reward of a bad paper exceeds the submission cost. If $c > \lambda r$, bad papers are never submitted, and the game trivially converges to perfect honesty.

3. **Reputation Constraint (Screening is preferred):**

$$DV_B(1 - \lambda) > BV_G\epsilon \tag{34}$$

   This condition ensures the journal prefers screening (OG) over blind acceptance (AA). It states that the aggregate penalty avoided by rejecting bad papers outweighs the benefit lost from accidentally rejecting good ones (Muller-Itten, 2022).

2.6.3 Stage 2: Conditional elimination (Economic dominance)

We now eliminate strategies based on the specific economic trade-offs defined above.

Dominance of Journal "Always Accept" (AA). We compare **Only Good (OG)** with **Always Accept (AA)**. Assuming the processing cost C_J applies to all submissions (Processing Cost Model), the difference depends on the net value of papers rejected by OG but accepted by AA:

$$\begin{aligned} U_J(\text{OG}) - U_J(\text{AA}) &= \bar{\rho}\left[(BM_G^{\text{OG}} - DM_B^{\text{OG}}) - (BM_G^{\text{AA}} - DM_B^{\text{AA}})\right] \\ &= \bar{\rho}\left[D(V_B - M_B^{\text{OG}}) - B(V_G - M_G^{\text{OG}})\right] \\ &= \bar{\rho}\left[DV_B(1-\lambda) - BV_G\epsilon\right] \end{aligned} \tag{35}$$

Under the **Reputation Constraint** (34), the penalty avoided (first term) exceeds the benefit lost (second term). Thus, **AA is strictly dominated by OG**. This result aligns with gatekeeper theory, suggesting that competitive journals maintain strict standards to maximize their distinctiveness and reputation (Muller-Itten, 2022).

Dominance of Journal "Reject All" (AR). Strategy AR yields a normalized payoff of 0 (or $-C_J V$). Since OG generates positive utility from good papers ($B > 0$) and filters bad ones, OG dominates AR provided the net value of the peer-accepted pool is positive: $BM_G > DM_B$. We assume the journal operates in a parameter region where valid science is net positive.

Dominance of Author "Never Submit" (NS). Comparing **Only Good (OG)** with **Never Submit (NS)** ($U_A = 0$):

$$U_A(\text{OG}, \text{OG}) - U_A(\text{NS}, \text{OG}) = \alpha\left[r\bar{\rho}(1-\epsilon) - c\right] \tag{36}$$

Under the **Viability Constraint** (32), this difference is positive as long as the congestion does not exceed the critical threshold $\bar{\rho} > \frac{c}{r(1-\epsilon)}$. In the viable regime, **NS is strictly dominated by OG** (and by AS). This condition reflects the rational participation constraint found in scientific contests, where authors only enter if the expected payoff exceeds the effort cost (Gross and Bergstrom, 2019).

2.6.4 The reduced evolutionary game

Following the two-stage elimination process, the complex multi-strategy interaction reduces to a concise 2×1 asymmetric game.

1. **The Journal** is a fixed institutional player locked into the **Only Good (OG)** strategy. It acts as the environment, applying a filter (ϵ, λ) and a capacity constraint K.
2. **The Author Population** evolves between two surviving pure strategies:
   - **Only Good (OG) [Cooperators]:** Authors who submit only high-quality work ($q = 1$).
   - **Always Submit (AS) [Defectors]:** Authors who submit everything ($q \in \{0, 1\}$), effectively spamming the journal with bad papers to exploit λ.

We define the **population state** $x(t) \in [0, 1]$ as the fraction of authors adopting the **Only Good** strategy. The evolutionary dynamics of the system are thus entirely determined by the time evolution of x. This reduction captures the essential dynamic of scientific evolution: the selection pressure between high-effort integrity and low-effort quantity (Smaldino and McElreath, 2016).

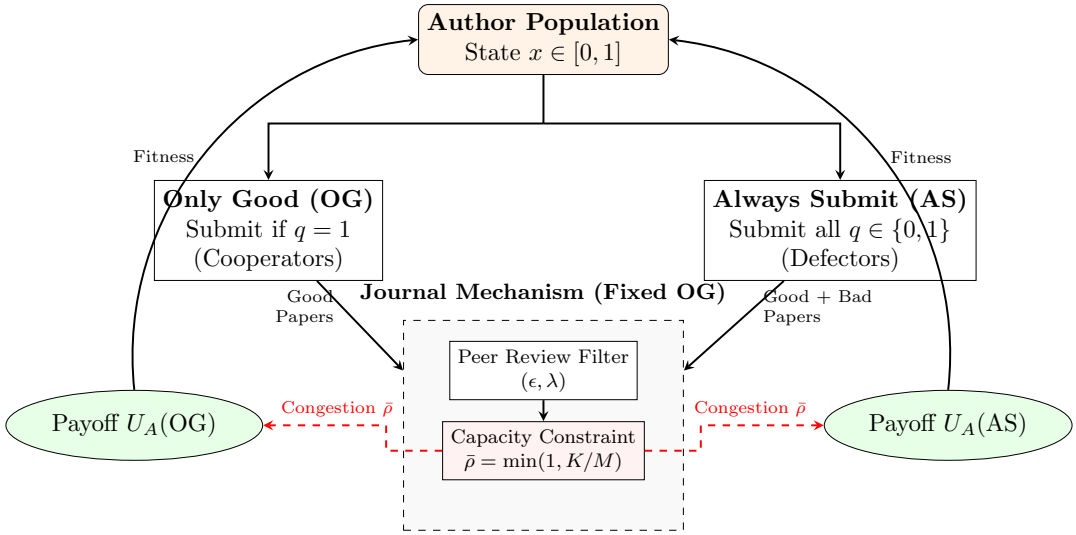


Figure 2: **Structure of the Reduced Evolutionary Game.** The author population evolves between two strategies: Only Good (OG) and Always Submit (AS). The Journal acts as a fixed mechanism consisting of a peer review filter and a capacity constraint. Submissions from AS authors increase the total volume M, triggering the capacity constraint. The resulting **Congestion Externality** ($\bar{\rho}$) reduces the publication probability for all papers, feeding back into the payoffs and driving the evolutionary dynamics.

3 Replicator Dynamics and Bifurcation Analysis

3.1 Model reduction and population state

Following the dominance analysis, the strategy space reduces to a dynamic contest between two surviving Author strategies, with the Journal playing a fixed filtering role. We define the **population state** $x(t) \in [0,1]$ as the proportion of authors playing **Only Good (OG)**, while $1 - x(t)$ play **Always Submit (AS)**. The evolutionary trajectory of this state is governed by the payoff differential between the strategies (Hofbauer and Sigmund, 1998).

We assume the system operates in the **Conditional Profitability Region**:

$$\lambda r < c < r \tag{37}$$

This condition implies that in the un-congested limit ($\eta \approx 1, \bar{\rho} \approx 1$), cheating is unprofitable ($\lambda r - c < 0$). Cheating becomes viable only if the screening mechanism collapses significantly due to congestion. This mirrors the "Market for Lemons" dynamic, where opportunistic behavior proliferates only when the buyer's ability to distinguish quality degrades (Akerlof, 1970; Thurner and Hanel, 2011).

3.2 Endogenous congestion functions

To capture the dual impact of congestion, we model two distinct bottlenecks. While numerical simulations use a smoothed approximation (Soft Minimum) to capture pre-congestion friction, we present the analytical forms using standard cutoffs for clarity.

3.2.1 Stage 1: The screening intensity function $\eta(x)$

The total submission volume is $V(x) = N[1 - (1 - \alpha)x]$. The screening intensity $\eta(x)$ represents the probability that a paper undergoes rigorous peer review. It degrades when volume exceeds the review capacity K_{rev}, a phenomenon consistent with models of "reviewer fatigue" and resource scarcity (Azar, 2005; Kovanis et al., 2016):

$$\eta(x) = \min\left(1, \frac{K_{rev}}{V(x)}\right) \tag{38}$$

We assume "rubber-stamping" implies unconditional acceptance of unscreened papers. This reflects the transition from rational selection to random selection observed in overloaded systems (Thurner and

Hanel, 2011). Thus, the pass probability for a bad paper, $\pi_0(x)$, increases as screening fails:

$$\pi_0(x) = \eta(x)\lambda + (1 - \eta(x)) \cdot 1 = 1 - \eta(x)(1 - \lambda) \tag{39}$$

For a good paper, the pass probability $\pi_1(x)$ also approaches 1 as scrutiny relaxes (assuming false negatives ϵ only occur during rigorous review):

$$\pi_1(x) = \eta(x)(1 - \epsilon) + (1 - \eta(x)) \cdot 1 = 1 - \eta(x)\epsilon \tag{40}$$

3.2.2 Stage 2: The rationing probability function $\bar{\rho}(x)$

Let K be the publication slot capacity. The expected volume of papers entering the "perceived good" pool $M(x)$ is:

$$\begin{aligned} M(x) &= V_G(x)\pi_1(x) + V_B(x)\pi_0(x) \\ &= N\alpha \cdot \pi_1(x) + N(1 - \alpha)(1 - x) \cdot \pi_0(x) \end{aligned} \tag{41}$$

Using a mean-field approximation, the expected rationing probability is:

$$\bar{\rho}(x) \approx \min\left(1, \frac{K}{M(x)}\right) \tag{42}$$

This formulation captures the "lottery nature" of high-impact publishing: when the number of scientifically sound papers (M) exceeds the journal's physical space (K), acceptance becomes a game of chance rather than pure merit (Casadevall and Fang, 2014).

3.3 Derivation of the replicator equation

Since both OG and AS strategies submit good papers whenever they occur (with identical acceptance probabilities), the expected payoff component from good papers cancels out in the differential. The dynamics are driven entirely by the net utility of the additional "bad" papers submitted by AS authors, denoted as $\Pi_{bad}(x)$. The replicator equation is:

$$\dot{x} = -x(1 - x)(1 - \alpha)\Pi_{bad}(x) \tag{43}$$

A bad paper yields reward r only if it passes both bottlenecks. The expected net utility is:

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c \tag{44}$$

3.4 Bifurcation and stability analysis

The system exhibits **Bistability** because the two congestion effects work in opposition, creating a dynamic tension characteristic of tipping-point models (Schelling, 1978):

- **Review Collapse ($\pi_0 \uparrow$):** Increases the incentive to cheat (pro-corruption).
- **Storage Rationing ($\bar{\rho} \downarrow$):** Decreases the incentive to cheat (pro-honesty).

When the review collapse effect dominates at low honesty, $\Pi_{bad}(x)$ becomes positive, creating a *corruption trap* where opportunistic strategies proliferate (Smaldino and McElreath, 2016). Conversely, when strict screening dominates at high honesty, $\Pi_{bad}(x)$ becomes negative, creating a basin of honesty.

3.4.1 Analytical Condition for Instability of the Internal Equilibrium

To rigorously characterize the stability of an interior equilibrium $x^* \in (0, 1)$, we analyze the local derivative of the velocity field. Following standard evolutionary stability analysis (Hofbauer and Sigmund, 1998), the equilibrium is unstable (a repeller) if and only if the net utility of cheating is locally decreasing in x:

$$\left.\frac{d\Pi_{bad}}{dx}\right|_{x^*} < 0 \tag{45}$$

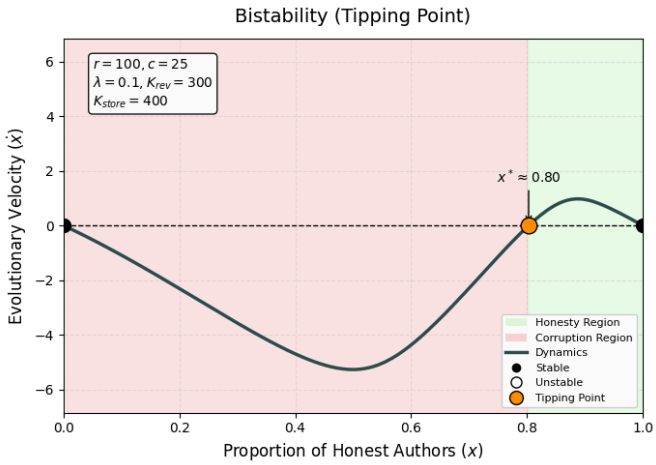


Figure 3: **Phase Portrait of the Dual-Bottleneck Dynamics (Bistability).** Parameters: $N = 1000, K_{rev} = 300, K_{store} = 400, r = 100, c = 25, \lambda = 0.1$. The curve shows the evolutionary velocity $\dot{x}$. The system exhibits three equilibria: (1) **Trap of Corruption** ($x = 0$): Stable ($\Pi_{bad}(0) > 0$). (2) **Basin of Honesty** ($x = 1$): Stable ($\Pi_{bad}(1) < 0$). (3) **Tipping Point** ($x^* \approx 0.82$): An unstable internal repeller. The orange marker indicates the critical threshold separating the two basins of attraction.

Recall that $\Pi_{bad}(x) = r \cdot \bar{\rho}(M(x)) \cdot \pi_0(x) - c$. Using the chain rule, we decompose the total derivative into two competing forces:

$$\frac{d\Pi_{bad}}{dx} = r \left(\underbrace{\bar{\rho}(M)\frac{d\pi_0}{dx}}_{\text{Review Effect}} + \underbrace{\pi_0(x)\frac{d\bar{\rho}}{dM}\frac{dM}{dx}}_{\text{Rationing Effect}} \right) \tag{46}$$

Note that $\frac{d\pi_0}{dx} < 0$ (since higher honesty reduces volume V, improving review quality η, thus lowering π_0). Conversely, the rationing term is typically positive (higher honesty reduces false positives in M, relaxing rationing, thus increasing $\bar{\rho}$). Therefore, the tipping point is unstable if and only if the Review-Dominated Condition holds locally:

$$\underbrace{\left|\bar{\rho}(M)\frac{d\pi_0}{dx}\right|}_{\text{Loss from stricter review}} > \underbrace{\pi_0(x)\frac{d\bar{\rho}}{dM}\frac{dM}{dx}}_{\text{Gain from relaxed rationing}} \tag{47}$$

This condition confirms that bistability arises when the recovery of peer review quality (as x increases) outweighs the relaxation of publication slots.

3.5 The regime of stable coexistence: Rationing-dominated dynamics

While the default scenario exhibits bistability, a fundamentally different topological regime emerges when the storage constraint (K_{store}) is significantly tighter than the review constraint (K_{rev}). Figure 4 illustrates this scenario, where the evolutionary velocity $\dot{x}$ transitions from positive to negative, creating a Stable Internal Equilibrium ($x^* \approx 0.44$).

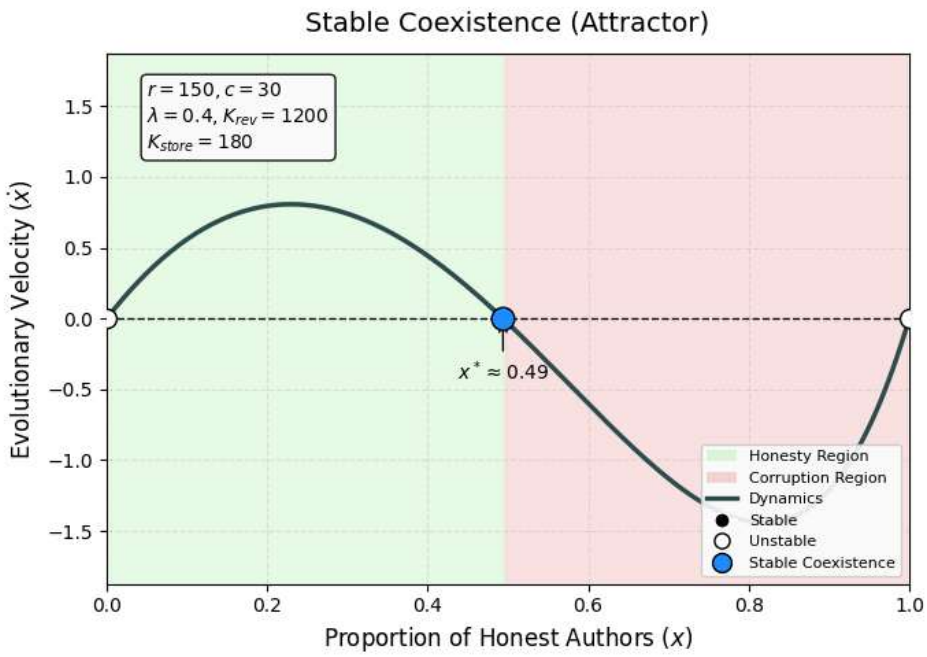


Figure 4: **Phase Portrait of Stable Coexistence (Rationing-Dominated).** Parameters: $N = 1000, K_{rev} = 1000, K_{store} = 180, r = 150, c = 30, \lambda = 0.4$. The system exhibits a stable attractor at $x^* \approx 0.49$. **Left Zone (Self-Correction):** At low honesty ($x < x^*$), severe storage rationing reduces the payoff below zero ($\Pi_{bad} < 0$), forcing the population to become more honest. **Right Zone (Temptation):** At high honesty ($x > x^*$), the relaxed rationing makes cheating profitable again ($\Pi_{bad} > 0$), forcing the population towards corruption. The system settles into a permanent mix of honest and opportunistic authors (Blue Point).

3.5.1 Existence and Stability Conditions

For a stable coexistence equilibrium to exist, two conditions must be met:

1. **Boundary Conditions:** The system must be repelled from both extremes. This requires $\Pi_{bad}(1) > 0$ (Temptation at Honesty) and $\Pi_{bad}(0) < 0$ (Unprofitability at Corruption). By the Intermediate Value Theorem, there must exist at least one root $x^* \in (0, 1)$ where $\Pi_{bad}(x^*) = 0$.

2. **Stability Condition:** The slope of the velocity field must be negative, i.e., $\frac{d\Pi_{bad}}{dx}\Big|_{x^*} > 0$. This corresponds to the Rationing-Dominated Condition*, where the "punishment" from overcrowding (tightening $\bar{\rho}$) outweighs the "opportunity" from collapsed screening.

3.5.2 Real-world implication: The "Defense by Congestion" paradox

The emergence of a stable internal equilibrium reveals a paradoxical phenomenon: **extreme congestion acts as an unintentional defense mechanism**. In journals with strict page limits ($K_{store} \ll N$), even if peer review fails ($\pi_0 \to 1$), the sheer volume of submissions drives the acceptance rate $\bar{\rho}$ so low that spamming becomes statistically unprofitable. This explains why top-tier journals do not degenerate into full corruption even under heavy load—the "lottery" is simply too hard to win. This dynamic aligns with contest models where excessive competition discourages entry by marginal participants (Gross and Bergstrom, 2019). However, it also implies vulnerability: if capacity is expanded (increasing K_{store}), the system loses this passive defense, potentially shifting the topology to bistability or collapse. This highlights the strategic value of artificial scarcity in maintaining the "gatekeeper" function of academic journals (Muller-Itten, 2022).

3.6 Boundary Regimes: Global Honesty and Global Corruption

Beyond the complex interior dynamics, the system may exhibit monotonic behaviors where the incentive to cheat is uniformly positive or negative.

3.6.1 Global Honesty ($x^* = 1$): Defense by Rigor

When $\Pi_{bad}(x) < 0$ for all $x \in [0, 1]$, the evolutionary velocity $\dot{x}$ remains positive throughout the domain, driving the system unconditionally toward $x^* = 1$ (Figure 5). A sufficient condition for this regime is that the payoff is negative even at the most favorable boundary ($x = 1$):

$$\Pi_{bad}(1) = r \cdot \lambda - c < 0 \tag{48}$$

This implies that suppressing the false-positive rate λ below the cost-reward ratio c/r is the primary safeguard for global honesty.

3.6.2 Global Corruption ($x^* = 0$): Systemic Failure

Conversely, when $\Pi_{bad}(x) > 0$ for all $x \in [0, 1]$, the system collapses unconditionally to $x^* = 0$ (Figure 6). This occurs when the payoff remains positive even under maximum congestion ($x = 0$):

$$\Pi_{bad}(0) = r \cdot \bar{\rho}(V_{max}) \cdot \pi_0(V_{max}) - c > 0 \tag{49}$$

This regime represents a total failure of governance, often seen in predatory journals where the profit motive drives acceptance regardless of quality.

3.7 Summary: A Topological Classification of Regimes

The dynamical behavior is ultimately determined by the interplay between the boundary payoffs. Table 3 summarizes the four topological regimes.

This classification reveals a dual-control mechanism: **Review Quality (λ)** secures the honest boundary ($x = 1$), while **Storage Capacity (K_{store})** secures the corrupt boundary ($x = 0$).

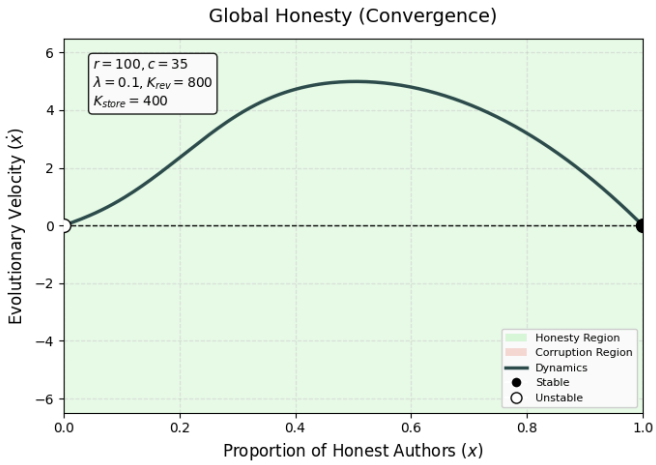


Figure 5: **Phase Portrait of Global Honesty.** Parameters: $N = 1000, K_{rev} = 800, K_{store} = 400, r = 100, c = 35, \lambda = 0.1$. The velocity is strictly positive (Green Zone), ensuring convergence to $x = 1$ from any initial state.

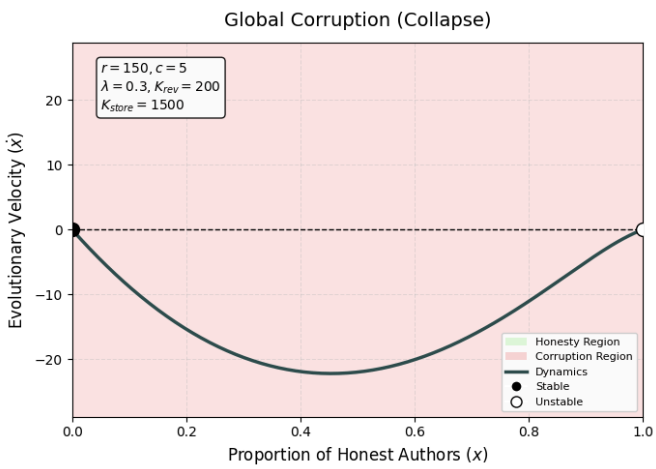


Figure 6: **Phase Portrait of Global Corruption.** Parameters: $N = 1000, K_{rev} = 200, K_{store} = 1500, r = 150, c = 5, \lambda = 0.3$. The velocity is strictly negative (Red Zone), ensuring collapse to $x = 0$.

Table 3: **Topological Classification of Dynamical Regimes based on Boundary Payoffs**

Regime	$\mathbf{\Pi_{bad}(1)}$ *(Low-spam Limit)*	$\mathbf{\Pi_{bad}(0)}$ *(Max-load Limit)*	**Dynamical Topology & Implication**
GH	< 0	< 0	**Honesty-dominant.** $x = 1$ is locally stable; $x = 0$ is unstable. Trajectories converge to $x = 1$ globally, *provided* $\Pi_{bad}(x)$ *has no interior root.*
GC	> 0	> 0	**Corruption-dominant.** $x = 0$ is locally stable; $x = 1$ is unstable. Trajectories collapse to $x = 0$ globally, *provided* $\Pi_{bad}(x)$ *has no interior root.*
BS	< 0	> 0	**Bistability.** Both boundaries are locally stable attractors. By continuity, **at least one** unstable interior equilibrium x^* exists to separate the basins.
SC	> 0	< 0	**Stable Coexistence.** Both boundaries are unstable repellers. By continuity, **at least one** stable interior equilibrium x^* exists (trapped by congestion). Uniqueness depends on monotonicity.

Note: The **Low-spam Limit** ($x = 1$) is driven primarily by review rigor (λ) via $\Pi_{bad}(1) \approx r\lambda - c$. The **Max-load Limit** ($x = 0$) is driven by the collapse of acceptance rate ($\bar{\rho}$) under maximum volume, regardless of review failure.

4 Parameter Sensitivity and Regime Transitions

4.1 Methodology and Baseline Calibration

The ultimate goal of any academic publishing policy is to maximize the **Average Publication Quality** (Q) and minimize the prevalence of academic misconduct. However, due to the non-linear interplay between the review bottleneck (η) and the storage bottleneck ($\bar{\rho}$), the system's response to policy changes is discontinuous. Instead of smooth adjustments, the ecosystem undergoes topological phase transitions between qualitatively distinct regimes.

To analyze these transitions, we employ **Numerical Bifurcation Analysis**. We define the instantaneous Average Publication Quality $Q(x)$ as the proportion of good papers within the set of accepted publications. This metric aligns with standard meta-scientific frameworks for assessing the reliability of research findings (Ioannidis, 2005):

$$Q(x) = \frac{N\alpha \cdot \pi_1(x)}{M(x)} = \frac{\alpha\pi_1(x)}{\alpha\pi_1(x) + (1-\alpha)(1-x)\pi_0(x)} \tag{50}$$

Crucially, the long-term quality Q_∞ is determined by the system's stable attractors. In bistable regimes, Q_∞ is path-dependent, meaning the final quality depends on whether the system started from a high-trust or low-trust state.

We classify the system into four regimes based on the stability of equilibria:

1. **Global Honesty (GH):** The system converges to $x = 1$, yielding $Q_\infty = 1.0$ (Perfect Quality).
2. **Global Corruption (GC):** The system collapses to $x = 0$, yielding $Q_\infty \approx \alpha$ (Baseline Quality). This state corresponds to a "Market for Lemons" where the journal fails to filter noise (Akerlof, 1970).
3. **Bistability (BS):** Two stable attractors exist. The outcome is determined by history (Hysteresis).
4. **Stable Coexistence (SC):** An internal equilibrium x^* exists, maintained by congestion. Quality stabilizes at a mediocre level ($Q_\infty \approx 0.5$).

To ensure comparative consistency, we establish two representative parameter sets (Table 4). **Set A (Baseline)** represents a typical "Vulnerable System" (e.g., a standard journal with moderate capacity),

prone to bistability. **Set B (High-Pressure)** represents a "Rationing-Dominated System" (e.g., a top-tier journal with strict page limits and high rewards), prone to stable coexistence.

Table 4: Parameter Calibration for Numerical Analysis

Symbol	Description	Set A (Baseline)	Set B (High-Pressure)
N	Total Population	1000	1000
α	Author Productivity	0.2	0.2
K_{rev}	Review Capacity	**300**	**1200**
K	Slot Capacity	**400**	**180**
r	Reward	100	**150**
c	Cost	25	**30**
λ	False Positive Rate	0.1	**0.4**
ϵ	False Negative Rate	0.05	0.01

Note: Set A is used as the default for sensitivity sweeps. Set B is used to illustrate the defense-by-congestion mechanism.

4.1.1 Justification for Dual Parameter Calibration

The necessity of defining two distinct parameter sets stems from the complex, non-monotonic nature of the dual-bottleneck mechanism. A single baseline cannot capture the full range of the system's dynamical behaviors. We distinguish between two fundamental ecosystem archetypes:

1. **Set A (The Vulnerable Regime - *Review-Dominated*):** This set represents the default state of many modern academic journals: moderate prestige ($r = 100$) and limited but not extreme storage constraints ($K = 400$). In this regime, the primary failure mode is the **Review Bottleneck** ($K_{rev} = 300$). When submission volume rises, the review process collapses before the storage limit is reached. This setup reflects the empirical reality of the "global burden" of peer review, where reviewer scarcity is the dominant constraint (Kovanis et al., 2016). This generates the **Bistable** topology, where the system is fragile and prone to catastrophic tipping.

2. **Set B (The High-Pressure Regime - *Rationing-Dominated*):** This set represents the ecosystem of top-tier prestige journals (e.g., *Nature* or *Science*). Here, the reward is significantly higher ($r = 150$), attracting massive attention. Crucially, the defining constraint is not the lack of reviewers (we assume ample review capacity $K_{rev} = 1200$), but the extreme scarcity of publication slots ($K = 180$). In this regime, the primary force is the **Storage Bottleneck**. This aligns with empirical findings on scientific gatekeeping, where high rejection rates are driven by physical slot limits rather than quality filtering alone (Siler et al., 2015; Casadevall and Fang, 2014). This generates the **Stable Coexistence** topology, illustrating the *Defense by Congestion* mechanism.

By analyzing these two distinct sets, we decouple the effects of *Review Collapse* versus *Slot Rationing*, ensuring our conclusions are robust across different segments of the academic publishing hierarchy.

4.2 The Decoupling of Capacities: Phase Diagram Analysis

In standard academic models, "journal capacity" is often treated as a unified resource. However, our dual-bottleneck framework reveals that the system's fate depends on the precise decoupling of Review Capacity (K_{rev}) and Slot Capacity (K_{store}).

To clarify the distinct roles of these parameters, we define the following key variables:

- K_{rev}: **Review Capacity**. Determines the screening intensity $\eta(x)$. High K_{rev} prevents review collapse.

- K_{store}: **Slot Capacity**. Determines the rationing probability $\bar{\rho}(x)$. Low K_{store} creates a "Defense by Congestion."

- $\eta(x)$: Probability of rigorous screening (Stage 1).

- $\bar{\rho}(x)$: Probability of publication after passing Stage 1 (Stage 2).
- $Q(x)$: Average Publication Quality, defined as the proportion of good papers among all accepted papers.

To visualize the interaction between these forces, we perform a two-dimensional bifurcation scan across the parameter space (K_{rev}, K_{store}). The resulting Phase Diagram (Figure 7) maps the system's topological regimes.

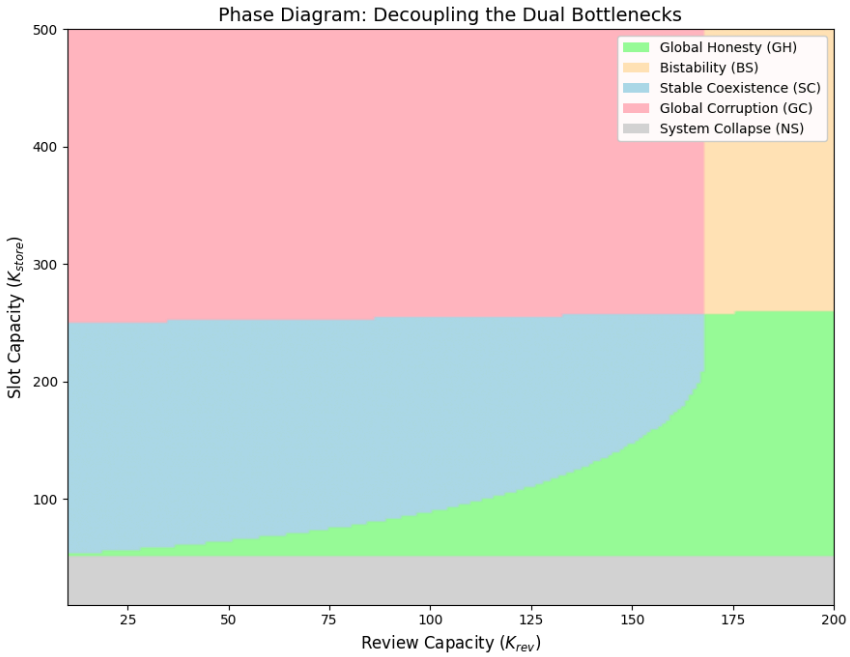


Figure 7: **Phase Diagram of the Dual-Bottleneck System.** The map classifies the topological regime based on the availability of review resources (K_{rev}) versus publication slots (K_{store}). **Green (Global Honesty):** High review capacity ensures cheating is unprofitable ($Q \to 1.0$). **Blue (Stable Coexistence):** Extreme slot scarcity ($K_{store} \ll N$) triggers the "Defense by Congestion," stabilizing the system at a mediocre but functional level. **Yellow (Bistability):** A transition zone where the outcome is history-dependent. **Red (Global Corruption):** When slots are abundant but review is weak, the defense mechanism vanishes, leading to total collapse ($Q \to \alpha$). **Gray (System Collapse):** Feasibility constraint violated (K_{store} too low); even honest authors cannot break even.

The topology of Figure 7 highlights two critical dynamical phenomena:

4.2.1 The Paradox of Slot Expansion

Moving vertically from the bottom of the diagram corresponds to increasing the journal's publication volume (K_{store}). The diagram reveals a non-monotonic response:

1. **From Gray to Blue (Feasibility):** Initially, increasing slots rescues the system from total collapse (Gray), making participation rational for honest authors.
2. **From Blue to Red (Destabilization):** However, further increasing K_{store} leads to a dangerous trajectory. Rising from the **Stable Coexistence (Blue)** zone, the system crosses into **Global Corruption (Red)** or **Bistability (Yellow)**.

This latter transition is the **Paradox of Slot Expansion**. It occurs because tight storage constraints act as a passive "Defense by Congestion," keeping $\bar{\rho}$ low enough to deter spam. By increasing K_{store}, the journal inadvertently dismantles this defense. As $\bar{\rho}$ rises, the "lottery" becomes winnable again. If K_{rev} is not increased simultaneously to catch the resulting flood of bad papers, the incentive to cheat reactivates,

and the system tips into full corruption. This finding aligns with strategic gatekeeping models, suggesting that while expanding capacity improves access, doing so without scaling review rigour destroys the signal value of the journal (Muller-Itten, 2022).

4.2.2 The Necessity of Review Resources

Moving horizontally to the right corresponds to investing in Review Capacity (K_{rev}). Provided the system is in a feasible region (not Gray), increasing K_{rev} is unconditionally beneficial, expanding the **Global Honesty (Green)** zone. Crucially, the boundary between corruption and honesty slopes upward: as K_{store} increases (more slots), the journal requires *disproportionately* more review capacity to maintain honesty. This implies that larger journals are inherently harder to police than smaller ones. They require a review infrastructure that scales super-linearly with volume to maintain the same level of integrity, exacerbating the already critical "global burden" of peer review (Kovanis et al., 2016).

4.3 Population Pressure and Hysteresis: The Cost of Lost Trust

While capacity parameters define the structural limits of the journal, the Total Population (N) represents the external load. A naive intuition assuming monotonic degradation might suggest that quality declines smoothly as load increases. However, our analysis reveals a critical **Hysteresis** phenomenon in the Bistable regime, indicating that the system's response to population pressure is highly path-dependent. This aligns with theories of catastrophic shifts in complex systems, where state recovery requires significantly more effort than state maintenance (Scheffer et al., 2001).

To rigorously capture this effect, we employ a numerical **Adiabatic Continuation** method. We define the system state $x_{st}(N)$ as the stable equilibrium selected by continuity from the previous state:

$$x_{st}(N_i) = \arg\min_{x \in \mathcal{E}_{st}(N_i)} |x - x_{st}(N_{i-1})| \tag{51}$$

where $\mathcal{E}_{st}(N_i)$ is the set of stable fixed points at N_i.

- **Forward Path (Crowding):** Increments N, initializing from $x_{st}(N_{min}) \approx 1$.
- **Backward Path (Recovery):** Decrements N, initializing from $x_{st}(N_{max}) \approx 0$.

Figure 8 illustrates the resulting trajectories using Baseline Set A.

The analysis yields three critical insights:

4.3.1 Structural Resilience (Forward Path)

A journal with a strong reputation demonstrates remarkable resilience. As seen in the Blue trajectory, the system maintains $x_{st} \approx 1$ and perfect quality ($Q = 1.0$) even when N far exceeds the nominal capacity. This persists because honest authors self-regulate, keeping the effective load $V(1) = N\alpha$ manageable. The **Collapse Point** $N_\uparrow \approx 1800$ is mathematically defined as the threshold where the honest equilibrium loses stability: $N_\uparrow = \inf\{N : \Pi_{bad}(1; N) \geq 0\}$.

4.3.2 The Stickiness of Corruption (Backward Path)

Once the system settles into the corrupt equilibrium, the sheer volume of spam saturates the review capacity. Even if N is reduced, the system remains trapped in the low-quality state as long as $\Pi_{bad}(0; N) > 0$. The **Recovery Point** $N_\downarrow \approx 350$ corresponds to the threshold where the corrupt equilibrium finally becomes unstable. This asymmetry mirrors the dynamics of collective reputation, where a group remains locked in a "bad equilibrium" long after external conditions improve, because the burden of proof is too high for individual reformers (Tirole, 1996).

Crucially, we define the Average Publication Quality $Q(x)$ as the proportion of good papers in the passed pool $M(x)$:

$$Q(x) = \frac{N\alpha\pi_1(x)}{N\alpha\pi_1(x) + N(1-\alpha)(1-x)\pi_0(x)} \tag{52}$$

Since slot rationing ($\bar{\rho}$) applies uniformly to $M(x)$, it does not alter the ratio of good-to-bad papers. Consequently, quality remains low throughout the backward path until the critical transition occurs.

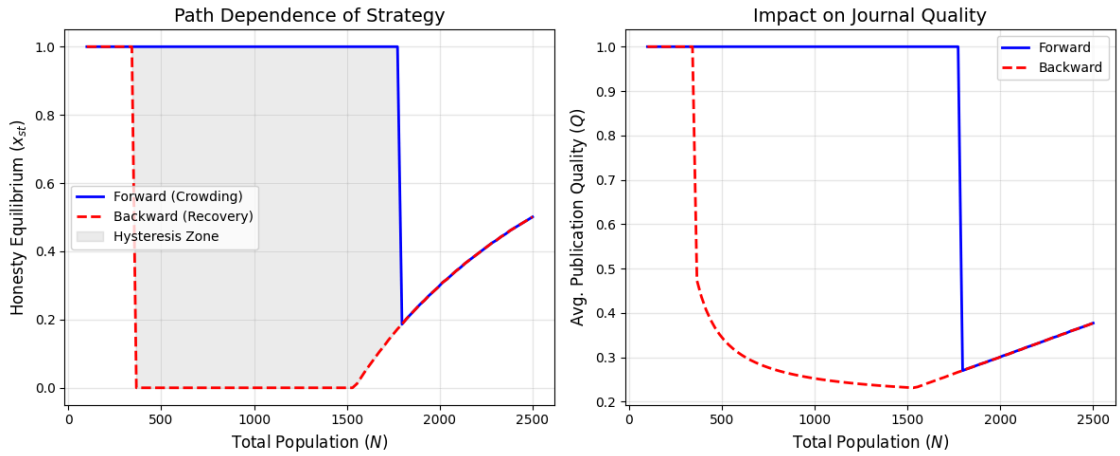


Figure 8: **Hysteresis Loop and Quality Collapse (Baseline Set A).** The shaded region represents the bistable interval $[N_\downarrow, N_\uparrow]$. **Left Panel (State):** On the Forward Path (Blue), the "Honest Norm" is resilient ($x_{st} \approx 1$) until the collapse threshold $N_\uparrow \approx 1800$. Once corrupted (Red Dashed), the system is trapped in the low-honesty attractor ($x_{st} \approx 0$). Recovery requires reducing N below the distinct recovery threshold $N_\downarrow \approx 350$. **Right Panel (Quality):** The collapse leads to a discrete jump in quality. In the corrupt state (Red Dashed), quality is dominated by the baseline productivity ($\alpha = 0.2$) but fluctuates slightly as review intensity $\eta(N)$ varies. Recovery only occurs when the system crosses $N_\downarrow$.

4.3.3 Policy Implication: Basin Hopping

The disparity between $N_\uparrow$ and $N_\downarrow$ implies that incremental reforms are structurally insufficient for a corrupted journal. Within the bistable interval, marginal reductions in submissions do not change the long-run attractor. Escaping the low-quality basin requires interventions strong enough to move the system across the recovery threshold ($N < N_\downarrow$). This suggests that while "soft" measures work for maintenance, "hard" measures (e.g., temporary submission moratoriums) are structural necessities for recovery.

4.4 Strategic Levers and Regime Transitions

While capacity parameters define the structural limits, strategic parameters $(r, c, \lambda, \epsilon, \alpha)$ determine the incentives. To provide a comprehensive policy guide, we perform a comparative sensitivity analysis across all five dimensions. The following analysis decomposes these levers into three functional groups: Screening, Economic Incentives, and Productivity.

4.4.1 Screening Parameters: The "Kill Switch" vs. The "Silent Factor"

First, we analyze the technical parameters controlling the peer review filter (Figure 9).

Panel (a) reveals that the **False Positive Rate (λ)** is the most critical parameter in the entire system. As λ increases, the system shifts rapidly from **Green (Honesty)** to **Yellow (Bistability)**, and finally to **Red (Corruption)**. In the yellow zone, resilience is lost—a minor perturbation can trigger

a catastrophic tip into the corrupt equilibrium. This aligns with signaling theory, where the reliability of the signal (low λ) is paramount for maintaining the separating equilibrium (Tiokhin et al., 2021).

In stark contrast, Panel (b) shows that **Friction (ϵ)** acts as a "Silent Factor." Within the explored range, the regime boundaries remain largely unchanged. This implies that policies focused solely on "efficiency" (reducing administrative friction for good papers) yield negligible structural improvements in integrity compared to improving detection accuracy.

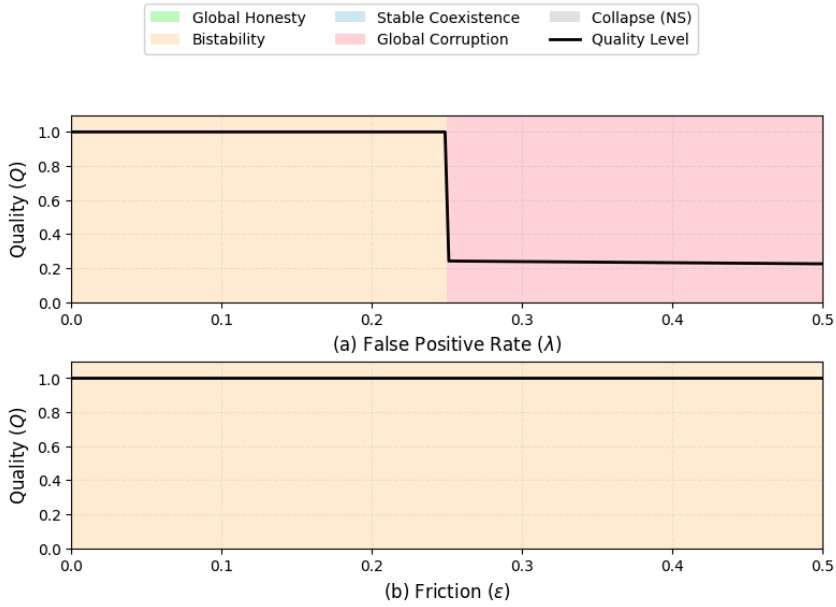


Figure 9: **Screening Parameters and Regime Stability. (a) False Positive Rate λ:** Acts as a major destabilizer. Increasing λ destroys the honest equilibrium, pushing the system into high-risk bistability. **(b) Friction ϵ:** Has negligible impact on regime boundaries, acting merely as a silent administrative cost.

4.4.2 Economic Levers: The Filter and The Accelerant

Next, we examine the financial and reputational incentives (Figure 10).

Panel (c) demonstrates that high **Rewards (r)** act as an accelerant for corruption. Increasing r significantly expands the **Red** and **Yellow** zones. When publication rewards are excessive, the "lottery" of rubber-stamping becomes profitable even at low probabilities. This confirms the mechanism of "natural selection of bad science," where high incentives select for strategies that maximize output at the expense of quality (Smaldino and McElreath, 2016).

Conversely, Panel (d) identifies **Submission Cost (c)** as a powerful filter. Increasing costs effectively shifts the system from **Red** back to **Green**. This provides theoretical support for submission fees as a tool to reduce congestion and filter out low-quality submissions (Cotton, 2013). However, pushing costs too high forces the system into the **Gray Zone**, where even honest authors cannot sustain participation ($\Pi_{good} < 0$).

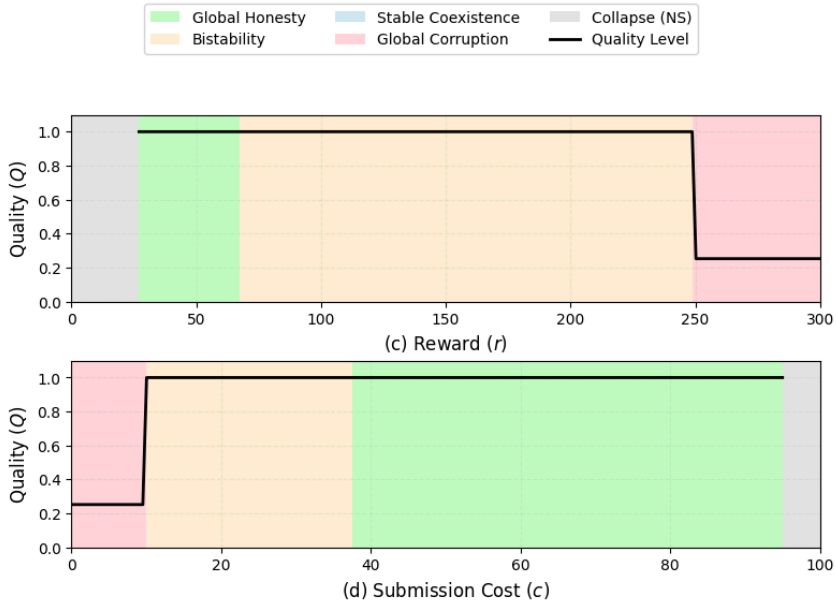


Figure 10: **Economic Incentives and Filtering. (c) Reward r:** High rewards expand the corrupt zones (Red/Yellow), incentivizing spamming strategies. **(d) Cost c:** Submission fees filter out spam, restoring Green, but excessive costs kill the journal (Gray).

4.4.3 The Productivity Paradox

Finally, we analyze the impact of aggregate author productivity (Figure 11).

Panel (e) highlights a counter-intuitive risk: as **Productivity (α)** increases, the system becomes *less* stable. Although the average quality (black line) rises mechanically due to a better input mix, the background topology shifts towards unstable regimes (Yellow/Red). This is the **Productivity Paradox**: as legitimate authors produce more papers, the total volume rises, consuming scarce review capacity. If K_{rev} is fixed, this "good congestion" dilutes the screening intensity for everyone, paradoxically making the system more vulnerable to invasion by spammers.

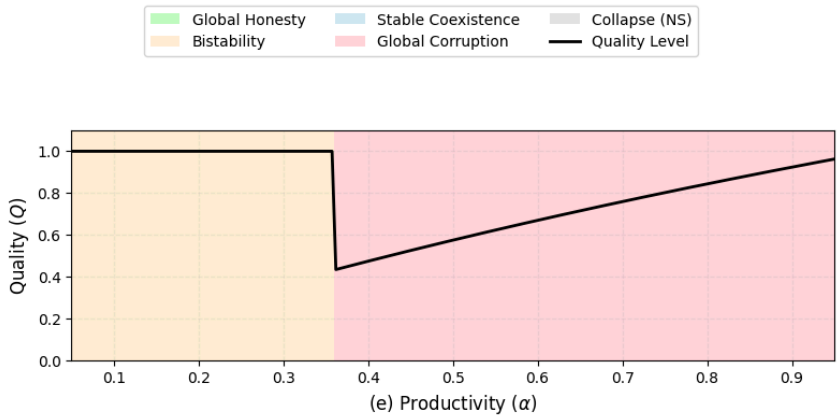


Figure 11: **The Productivity Paradox. (e) Productivity α:** Higher productivity increases total congestion. While observable quality rises (black line), the structural stability degrades (background shifts to Red), narrowing the window of stability.

5 Agent-Based Modeling

5.1 Motivation and Micro-foundations

The analytical results presented in previous sections rely on the replicator dynamics, which assume an infinite population ($N \to \infty$) and a deterministic evolution of strategies Hofbauer and Sigmund (1998). However, real-world academic communities consist of a finite number of researchers subject to stochastic fluctuations and finite-size effects Nowak (2006). To validate the robustness of our topological classification and the mathematical dominance analysis, we implement a microscopic **Agent-Based Model (ABM)**. This approach, widely used in the study of complex adaptive systems Bonabeau (2002), serves three critical purposes: first, it tests whether the strictly dominated strategies naturally die out in a dynamic environment; second, it verifies if the predicted equilibria emerge from local interactions under stochastic noise; third, it confirms that the "Defense by Congestion" mechanism is not an artifact of the mean-field approximation.

5.2 Simulation Design

We simulate a population of $N = 1000$ agents. The time evolution follows the standard protocols of stochastic evolutionary game theory in finite populations Traulsen and Hauert (2009). To rigorously test the system's evolutionary trajectory without assuming a pre-reduced state, the population is initialized with a uniform distribution across all four strategies (25% for each strategy). The process proceeds as follows:

1. **Interaction Phase:** In each time step t, agents generate submissions based on their full strategy space ($s_i \in \{\text{OG}, \text{AS}, \text{OB}, \text{NS}\}$). The aggregate volume V and peer-accepted volume M are calculated endogenously, determining the congestion levels η and ρ.

2. **Payoff Realization:** Agents receive payoffs Π_i based on the realized publication outcomes. This captures the "lottery" nature of the congestion bottleneck.

3. **Strategy Update (Social Learning):** We employ the **pairwise comparison process** to model strategy evolution. A randomly selected agent A considers mimicking agent B's strategy. The probability of switching $s_A \to s_B$ is given by the Fermi function derived from statistical mechanics Szabó and Tőke (1998):

$$P(s_A \to s_B) = \frac{1}{1 + \exp(-\beta[\Pi_B - \Pi_A])} \tag{53}$$

where β represents the intensity of selection. This update rule is the standard method for introducing bounded rationality and decision noise into evolutionary dynamics Traulsen and Hauert (2009).

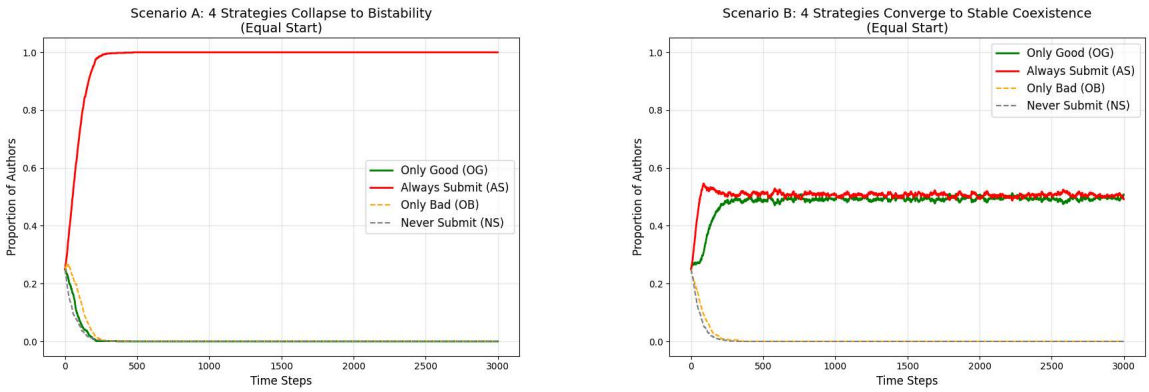
5.3 Simulation Results

We performed simulations using the parameter sets defined in Table 4, employing a high selection intensity ($\beta = 1.0$) to reduce noise. As shown in Figure 12, the microscopic agent-based simulations provide strong corroboration for both the strategy elimination and the macroscopic predictions derived in Section 3.

Validation of Dominance Analysis: In both scenarios, the "Only Bad" (OB) and "Never Submit" (NS) strategies are rapidly driven to extinction within the first few hundred time steps. This dynamic directly verifies the Iterated Elimination of Strictly Dominated Strategies (IESDS) conducted in Section 2.6. It demonstrates that the system autonomously reduces to a binary contest between OG and AS, aligning with evolutionary game theory findings that strictly dominated strategies fail to survive under monotonic selection dynamics Samuelson (1997).

Scenario A (Review-Dominated): In the review-dominated regime, the dynamics exhibit sharp bistability (Figure 12a). Because the initial proportion of honest OG authors (25%) falls well below the theoretical unstable tipping point ($x^* \approx 0.82$), the system is unable to self-correct. Once the inferior OB and NS strategies vanish, the AS strategy outcompetes the remaining honest authors, driving the system into the absorbing state of full corruption ($x = 0$). This confirms the severe path-dependence and "Hysteresis" trap inherent in complex systems with alternative stable states Scheffer et al. (2001); Schelling (1978).

Scenario B (Rationing-Dominated): In the rationing-dominated regime, the population converges to a stable interior equilibrium (Figure 12b). Despite the initial chaotic interaction of all four strategies, the population rapidly settles to a long-run average of $x^* \approx 0.49$ for OG and AS, while OB and NS remain extinct. This result aligns precisely with the analytical fixed point ($x^* \approx 0.49$) identified in the phase portrait analysis. The close correspondence confirms that the **"Defense by Congestion"** mechanism is a robust emergent property, immune to initial strategy turbulence and finite-size stochasticity.



(a) Scenario A: Review-Dominated Dynamics

(b) Scenario B: Stable Coexistence

Figure 12: **Agent-Based Model Trajectories with 4 Strategies ($\beta = 1.0$).** The plots show the time evolution of the strategy proportions. **(a)** Strong Bistability: Starting from an equal uniform mix, the system rapidly eliminates OB and NS. Because the initial OG fraction (0.25) is below the critical tipping point, the system collapses to full corruption ($x = 0$). **(b)** Stable Coexistence: OB and NS are quickly eliminated, and the remaining OG and AS populations converge to a mixed equilibrium ($x^* \approx 0.49$). The precise agreement with analytical predictions confirms both the theoretical dominance analysis and the structural congestion stabilization.

5.4 Co-operation ABM

To rigorously validate the robustness of our theoretical findings and explore the microscopic foundations of the "Defense by Congestion" mechanism, we develop a specialized Co-operation Agent-Based Model (ABM) featuring local batch-matching. While the standard replicator dynamics rely on a mean-field approximation assuming an infinite population and a single monolithic market, real-world academic publishing is inherently fragmented. Authors typically compete within specific sub-disciplines, specialized journals, or distinct temporal submission windows. The primary motivation for designing this ABM is to investigate whether the macroscopic stability predicted by the ODEs holds when subjected to the localized uncertainty, heterogeneous payoffs, and stochastic fluctuations characteristic of finite populations.

The core innovation of this extended ABM is the *batch-matching* algorithm. Instead of evaluating all N agents globally, the population is randomly partitioned into local batches (sub-markets) in each time step. The journal's review and storage capacities (K_{rev} and K_{store}) are scaled proportionally to match these local batches. This approach significantly improves upon standard global ABMs by introducing critical variance into the localized rationing probability (ρ) and enabling stochastic drift. It captures the granular reality that the publication "lottery" is not uniform; different sub-fields experience varying degrees of congestion simultaneously. This localized competition provides a much more rigorous stress-test for the theoretical equilibria.

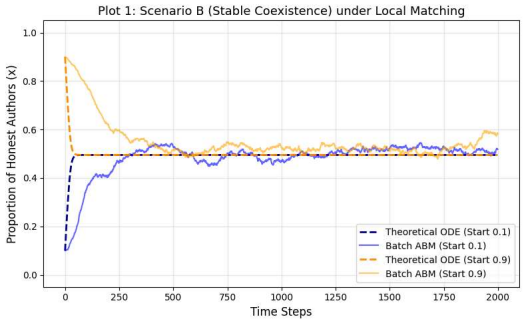
The simulation results validate our threshold competition framework across four key dimensions, as illustrated in Figure 13:

Trajectory Analysis and Robustness: Figures 13(a) and 13(b) illustrate the time evolution

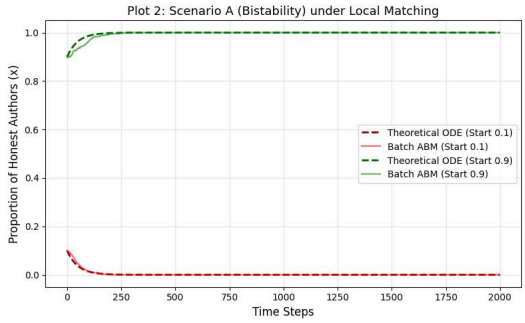
of strategies under batch matching. In the rationing-dominated regime (Scenario B), despite chaotic local interactions, both the theoretical ODE and the ABM trajectories converge tightly to the stable internal equilibrium of $x^* \approx 0.49$. This confirms that the stable coexistence induced by congestion is highly robust against local noise. Conversely, in the review-dominated regime (Scenario A), the ABM demonstrates sharp bistability, collapsing rapidly to global honesty ($x = 1.0$) or global corruption ($x = 0.0$) depending on initial conditions. The ABM closely traces the expected theoretical paths but exhibits micro-oscillations, validating the IESDS reduction while underscoring the severe path-dependence of the bistable regime.

Distribution of Local Congestion: Figure 13(c) provides the empirical distribution of the realized publication probability (ρ) across all local batches during the steady state. The mean-field model assumes a singular, fixed expected ρ. However, the batch ABM reveals a distribution clustered around a mean of $\rho = 0.51$. This finding is conceptually crucial: it demonstrates that the system-wide "Defense by Congestion" does not require perfectly uniform bottleneck pressure. Macroscopic integrity is maintained through the statistical consistency of localized congestion, aligning with models of decentralized and imperfect scientific evaluation (Thurner and Hanel, 2011).

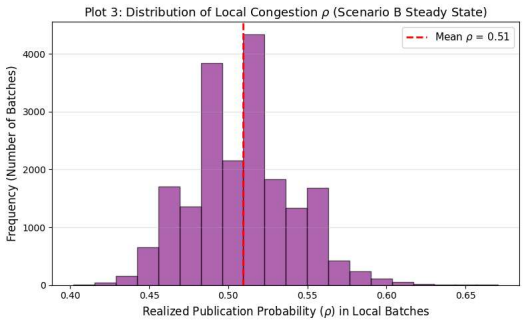
Phase Portrait and Stochastic Drift: Finally, Figure 13(d) maps the evolutionary phase portrait, superimposing realized ABM stochastic drift (Δx) over the theoretical expected velocity curve. This visualization highlights a critical systemic vulnerability. Near the unstable tipping point ($x \approx 0.82$), the localized variance causes the realized Δx to scatter widely and frequently cross the zero-axis. This demonstrates that in bistable regimes, localized sequences of "bad luck" or noise in review batches can prematurely trigger a catastrophic shift into the corruption basin (Scheffer et al., 2001). It powerfully illustrates the fragility of scientific ecosystems, where micro-level stochasticity can amplify the natural selection of bad science if the macro-level tipping point is approached (Smaldino and McElreath, 2016).



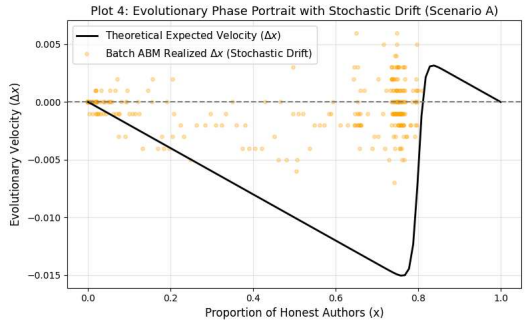
(a) Scenario B (Stable Coexistence) Trajectories



(b) Scenario A (Bistability) Trajectories



(c) Distribution of Local Congestion (ρ)



(d) Phase Portrait with Stochastic Drift

Figure 13: **Results of the Co-operation ABM with Local Batch-Matching.** The composite figure demonstrates that the theoretical mean-field predictions hold even under the rigorous localized constraints and stochastic noise introduced by fragmented sub-markets. Notably, local variance (c) creates significant stochastic drift near unstable tipping points (d).

5.4.1 The Journal as an Independent Evolutionary Agent

From Fixed Environment to Co-Evolutionary Agent In previous sections, including the local batch-matching analysis in Section 5.4, the journal was mathematically treated as a fixed institutional player locked into the "Only Good" (OG) strategy. It acted purely as a static environmental constraint enforcing a peer review filter and a capacity limit. By elevating the journal to an independent, adaptive agent, we transition the framework from a simple population game into an asymmetric co-evolutionary game. Journals now possess their own complete strategy space, consisting of Only Good (OG), Always Accept (AA), Only Bad (OB), and Reject All (AR). They update their behavior based on their own distinct utility function, which must balance the reputation benefits of publishing good papers, the penalties of publishing bad papers, and the administrative costs of peer review. This structural update captures the "Red Queen" dynamic inherent in academic publishing: if the author population becomes overwhelmingly honest, journals are financially incentivized to drop their costly screening processes and adopt the "Always Accept" (AA) strategy. However, this relaxation immediately triggers an opportunistic invasion of "Always Submit" (AS) authors, which in turn forces the journals to revert to strict filtering.

The Significance of the 10:1 Local Batch-Matching Mechanism Real-world academic publishing is not a monolithic global market, but rather a highly fragmented ecosystem divided into specific sub-disciplines and specialized journals. To accurately model this micro-level reality, we introduce an extreme 10:1 local batch-matching mechanism, pairing 1000 authors with 100 distinct journal agents. In this configuration, global capacities (K_{rev} and K_{store}) are scaled down proportionally to match the size of these local sub-markets. The primary significance of this 10:1 ratio is the maximization of localized variance and stochastic drift. By breaking the assumption of uniform global congestion, a single journal agent shifting to a rubber-stamping (AA) strategy creates a massive, localized payoff spike for the specific batch of 10 authors assigned to it, severely testing the structural resilience of the system.

ABM Configuration and Result Analysis To simulate this environment, we employ an Agent-Based Model (ABM) where both populations update their strategies via a Fermi pairwise comparison process, introducing bounded rationality and decision noise.

First, we analyze the global co-evolutionary baseline without fragmentation, as illustrated in **Figure 14**. The ABM realization demonstrates that structurally dominated strategies, such as Only Bad (OB) and Reject All (AR), are rapidly driven to extinction. The remaining strategies enter a dynamic coexistence where the continuous mathematical predictions (ODE) and the finite-population ABM trajectories align closely.

However, under the extreme fragmentation of the 10:1 batch-matching architecture, illustrated in **Figure 15**, the system exhibits a profound deviation from the global mean-field predictions. In the rationing-dominated regime (Scenario B), the ABM reveals that the "Never Submit" (NS) strategy unexpectedly overtakes the ecosystem. This occurs due to a phenomenon we term "Feasibility Collapse." Because the global publication slots (K_{store}) are strictly divided among the 100 journals, each local sub-market operates with an extremely narrow bottleneck. If even a small number of opportunistic (AS) authors flood a specific 10-person room, the local rationing probability (ρ) plummets below the economic break-even threshold required to justify the submission cost (c). Consequently, the "Defense by Congestion" becomes too punitive; even honest authors (OG) suffer negative expected payoffs and are forced to exit the market entirely, leaving NS as the only rational, risk-averse strategy.

6 Conclusion

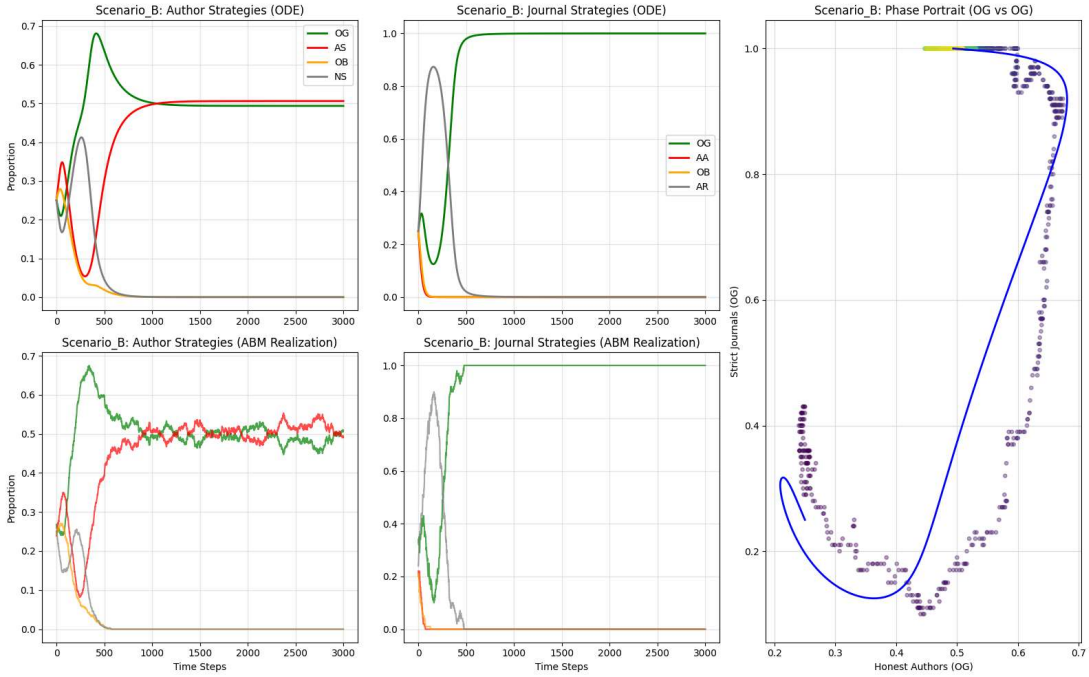


Figure 14: Global Co-evolutionary Baseline (Scenario B). The system demonstrates stable coexistence without fragmentation, where the finite-population ABM trajectory closely aligns with the continuous ODE predictions.

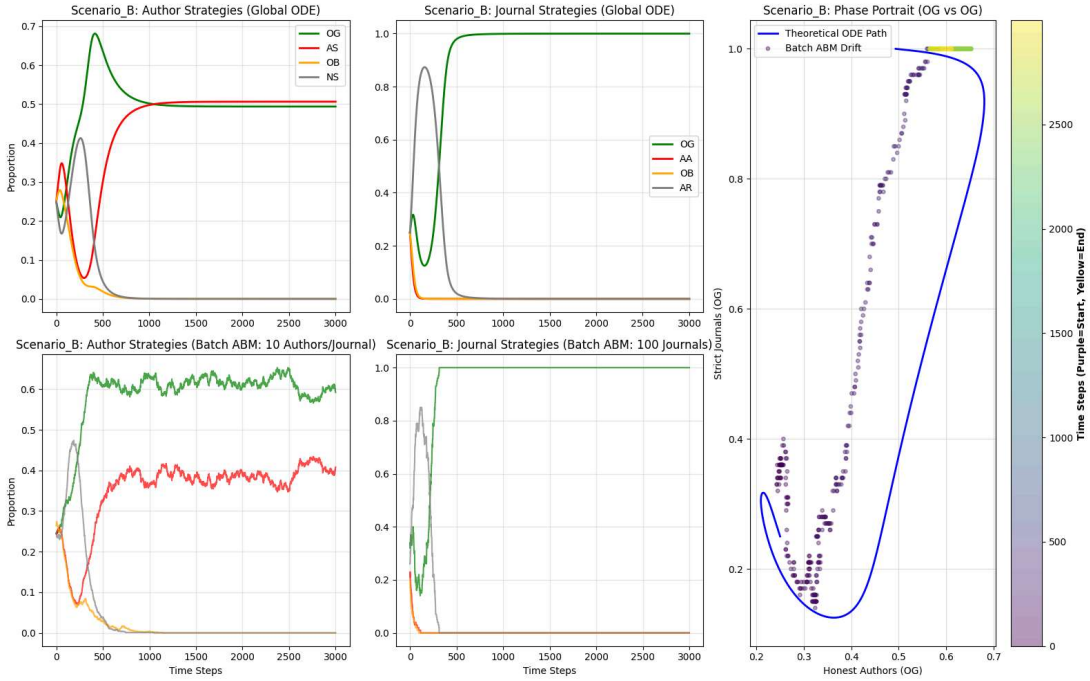


Figure 15: Co-evolutionary Dynamics under 10:1 Local Batch-Matching (Scenario B). The extreme fragmentation induces severe localized congestion variance, causing the "Never Submit" (NS) strategy to dominate due to Feasibility Collapse.

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